

EDTA solution		citric acid solution	
DI water	80–90 mL	DI water	80–90 mL
EDTA- Na_2	3.7224 g	citric acid	3.8424 g
add DI water until 100 mL is reached			
adjust pH to 7–9 with NaOH and H_2SO_4			

Table 4.5: **Iron chelation** solution recipes for a 0.1 mol L^{-1} EDTA solution and a 0.2 mol L^{-1} citric acid solution. Mix each solution until completely dissolved.

radiation.

III: Gallic acid swelling — swell ISO3 samples in gallic acid until fully swollen. Then dry and then UV-etch them at (254 nm, 50 mW cm^{-2} , for 2 h). Swelling and drying prior to etching should prevent crystallisation of gallic acid on the surface and ensure penetration throughout the matrix. But this may also cause swelling of the PEO and contribute to drying artefacts.

IV: Gallic acid and acetonitrile — soak ISO3 in a mixture of gallic acid and acetonitrile for 20 min, then dry and UV etch. Acetonitrile is intended to allow the PEO block to deswell and prevents problems that might occur in the etching suggestion **III**. However, this method may cause gallic acid to recrystallise, producing internal stress.

4.7 Distortion Analysis

The assessment and control of distortion in block copolymer self-assembly samples is of considerable importance, as structural deviations affect material properties and device functionality. A variety of advanced characterisation techniques are available to quantify such deviations, including atomic force microscopy (AFM) [40, 165], X-ray nanotomography [41, 97], grazing-incidence small-angle X-ray scattering (GISAXS) [166, 167], focused ion beam (FIB) scanning electron microscopy (SEM) cross-sectioning [40], FIB SEM tomography [93, 168], and transmission electron microscopy (TEM) cross-sections [169], as well as TEM tomography [170–172]. AFM is limited to the top surface, while FIB SEM and TEM cross-sections are limited to a 2D view of cross sections. Tomographic techniques, including X-ray, FIB SEM and TEM, offer deeper structural insight at the expense of significant time investment for both measurement and evaluation, particularly in the case of challenging samples.

In this work, an algorithm was developed to facilitate quantitative analysis of deformation, specifically targeting orientation, scaling and shear in block copolymer domains. This algorithm operates on four to six FIB SEM cross-sections obtained from a single grain (see Fig. 4.8a), an approach made feasible by the large domain sizes achieved in Chapter 6. Although the methodology showed promise, results lacked

sufficient conclusiveness to allow direct extraction of deformation parameters. Consequently, a manual alignment of cross-sections to a theoretical gyroid model, as described in Section 6.2.1, was employed. The algorithmic approach and its implementation in Python 3.13 are outlined below, anticipated to serve as a foundation for future developments in automated distortion analysis.

Algorithm workflow:

1. Four FIB SEM cross-sectional images, identified as originating from the same domain (Fig. 4.8g), are imported using `image.open()`, and cropped to a uniform size. For testing the algorithm, these cross-sections can also be generated (Fig. 4.8a).
2. Estimations are made for the unit cell size of the experimental data, and fill fraction. Presettings define the model size and model unit cell size in pixels for the correlation, the wobbling parameters for the iteration, and the number of iterations for the algorithm to stop.
3. A model meshgrid, is generated, according to the input of the meshgrid size and the unit cell, ranging from $-i \cdot \pi$ to $+i \cdot \pi$ for x, y and z coordinates, with i the number of unitcells in the meshgrid. The meshgrid gets transformed using random affine transformation matrices for rotation (M_{rot}), scaling (M_{scale}), and shear (M_{shear}). The combined deformation is expressed as:

$$M_{\text{deform}} = M_{\text{shear}} \cdot M_{\text{rot}} \cdot M_{\text{scale}}, \quad (4.4)$$

with

$$M_{\text{shear}} = \begin{bmatrix} 1 & sh_{xy} & sh_{xz} & 0 \\ sh_{yx} & 1 & sh_{yz} & 0 \\ sh_{zx} & sh_{zy} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (4.5)$$

$$M_{\text{rot}} = R_x \cdot R_y \cdot R_z, \quad (4.6)$$

$$R_x = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\alpha) & -\sin(\alpha) & 0 \\ 0 & \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (4.7)$$

$$R_y = \begin{bmatrix} \cos(\beta) & 0 & \sin \beta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(\beta) & 0 & \cos \beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (4.8)$$

$$R_z = \begin{bmatrix} \cos(\gamma) & -\sin(\gamma) & 0 & 0 \\ \sin(\gamma) & \cos(\gamma) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (4.9)$$

and

$$M_{\text{scale}} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4.10)$$

4. For each cross-section i , the transformed model m is sliced using a rotation matrix $M_{\text{rot},i}$:

$$G_{m,i} = M_{\text{rot},i} \cdot M_{\text{deform}} \cdot G_m \quad (4.11)$$

5. The deformed and rotated level-sets get binarised by applying the level-set equation on the X, Y and Z coordinates together with the fill fraction defining threshold t . The resulting four model cross-sections are labelled with the normal vector of the corresponding slicing plane.

$$t \lesseqgtr \sin(Lx) \cos(Ly) + \sin(Ly) \cos(Lz) + \cos(Lx) \sin(Lz). \quad (4.12)$$

6. Each simulated model cross-section is matched to its experimental counterpart using `cv2.matchTemplate` (serial Python implementation) or `conv2d` (in parallel using Pytorch). The spatial offset yielding the maximum correlation score is recorded.
7. An aggregate correlation score from the correlations with the cross-sections in front (F) and from the sides (S, either right or left) is calculated, representing overall alignment quality (Fig. 4.8c). The correlation score of each set of deformation parameters is mapped to a point on the rotation sphere (Fig. 4.8b).

$$C = \frac{1}{\sqrt{4}} \sqrt{C_{\text{S90}}^2 + C_{\text{S24}}^2 + C_{\text{F90}}^2 + C_{\text{F24}}^2}. \quad (4.13)$$

8. The procedure iterates, perturbing the deformation parameters (rotation, scaling, shear) to further improve the correlation score. Parameter sets yielding higher correlations supersede the existing optimum.

The codebase was initially implemented for single-core serial execution (Spyder, Python 3.13), and subsequently optimised for parallel computation using Pytorch on a high-performance CPU. Preliminary efforts towards GPU acceleration via Pytorch's `cuda` backend were initiated but not completed owing to time constraints. The principal differences between serial and parallel implementations are summarised in Table 4.6.

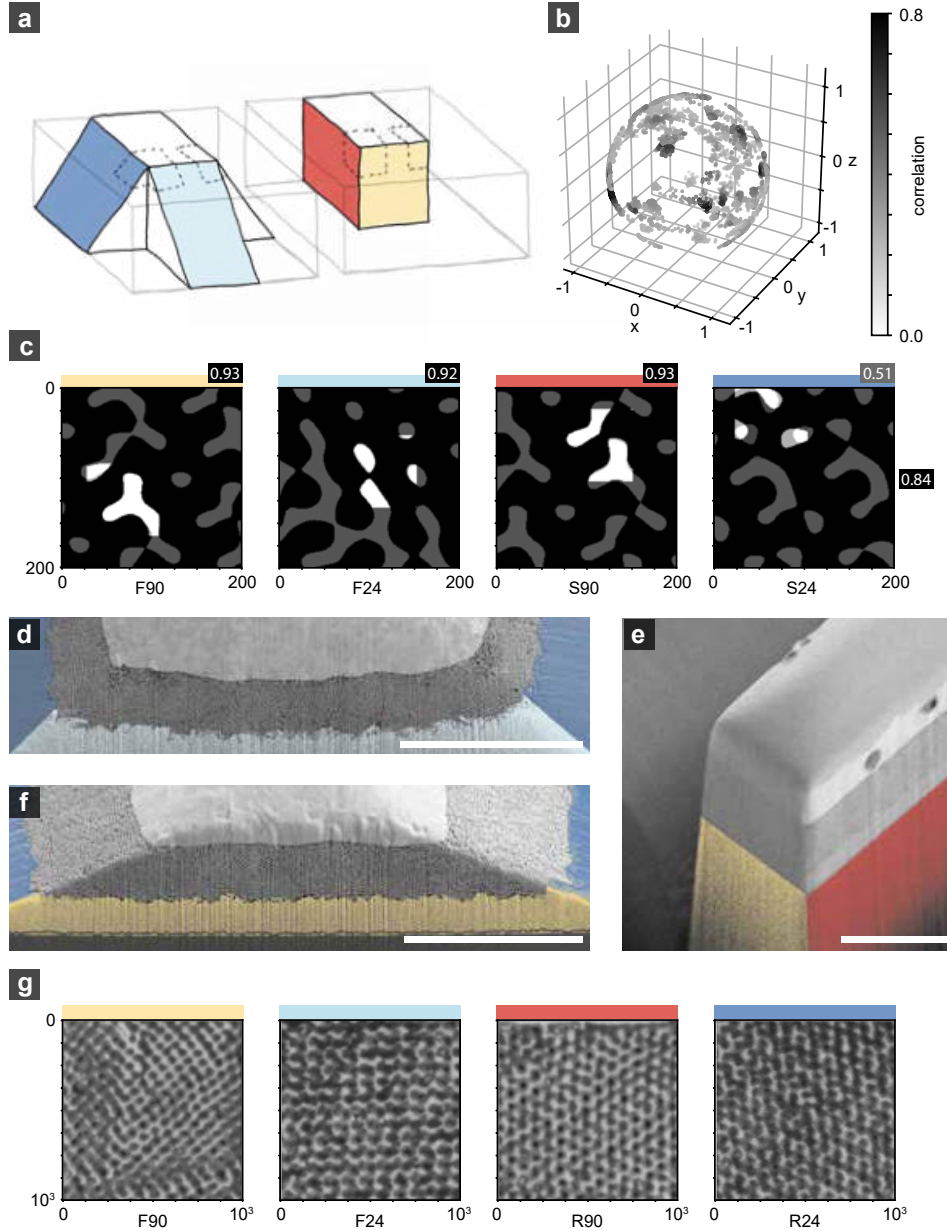


Figure 4.8: **Distortion analysis** on model cross-sections and on samples. **a** A $N \times N$ matrix is rotated and cut with angles of 24° (cold shades) and angles of 90° (warm shades) with respect to the top surface. Either the left or right corner of the cube is taken for the four cross-sections. **c** Two cross-sections on the model at the front of the cube (F90, F24), a cross-section from the side (S90) and one that is not matching with the others (S24) in the background, shaded in gray. The smaller bright square is the probing cross-section from the code, finding good correlation among the first three cross-sections and a mediocre one for the fourth (S24) with an overall correlation of 0.84. **b** The correlation over all four cross-sections in dependence of the top-surface orientation can be displayed in a scatterplot on the surface of the sphere. **d** FIB-SEM cross-section overview on 24° -, **e** 90° -, and **f** a mixture of 90° - and 24° -cross-sections. **g** Selected cross-sections at the right corner of a sample. Scale bars are **d** $4\mu\text{m}$, **e** $2\mu\text{m}$, **f** $3\mu\text{m}$.

Type	Single	Parallel
level-set	matrices	float tensors (torch)
correlation	cv2.matchTemplate	conv2D
device selection	-	'cpu'/'cuda'

Table 4.6: **Python vs. Pytorch**

Limitations and outlook: The algorithm was able to optimise the deformation parameters of the model meshgrid in both serial and parallel implementations. Nonetheless, the optimum correlation value did not stand out significantly compared to other local maxima, with high correlation scores often resulting from strong agreement with only a single cross-section. Extended computational runs or alternative algorithmic strategies are therefore required to yield a more robust total correlation.

Future improvements should focus on three main aspects:

1. **Image pre-processing** - further enhancement by maximising contrast or ensuring consistent binarisation across all experimental cross-sections, while preserving critical structural details, could improve matching accuracy.
2. **GPU acceleration** - migrating the computation to a graphics processing unit is expected to markedly reduce run times and may clarify whether current limitations in correlation are artefacts of computational constraints.
3. **Depth emulation** - as FIB SEM imaging inherently samples a finite volume rather than an idealised plane, a more representative comparison could be obtained by summing and normalising several adjacent model cross-sections, thereby appropriately emulating the effective imaging depth.

The latest version of the files used for the distortion analysis are uploaded on GitHub:

https://github.com/iselire/Distortion_Analysis_TPMS.

